AXLE JOURNAL

OIL LUBRICATED STRAIGHT ROLLER BEARING

The minimum axle journal diameter is governed by the interference fit between the axle journal and the bearing inner race.

A new axle journal diameter is 6.878" to 6.879" (6.503" to 6.504" on 6-1/2" journal bearing axle). This provides a minimum interference fit of .003" to the inner race.

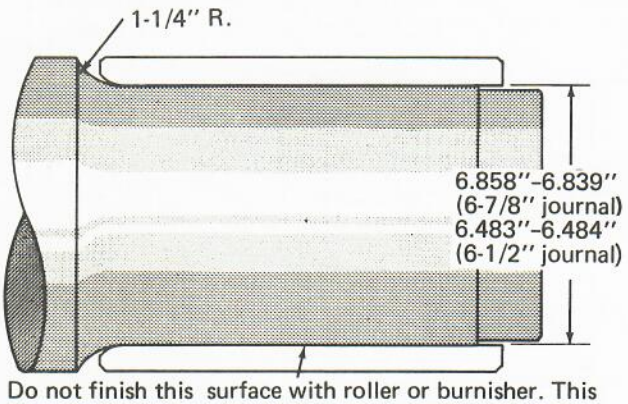
A service limit of 6.8765" (6.5015" for 6-1/2" journal bearing axle) minimum axle journal diameter provides an interchangeable interference fit of .0015". If it is desired to use a selective fit, an axle journal diameter of 6.8755" (6.5005" for 6-1/2" journal bearing axle) can be used with an inner race having a minimum I.D. of 6.8740" (6.499" on 6-1/2" journal bearing axle) to obtain the .0015".

The minimum interference fit of .0015" for any axle, is only permitted providing that:

1. Inner race does not show fretting in excess of 20% of total I.D. surface.

2. Axle journal surface does not show fretting in excess of 1" wide or 10% of the total surface.

If the .0015" interference fit cannot be obtained with a standard inner race, the axle journal can be machined to the dimensions shown in Fig. 26. However, the interference fit should be .003" to .005".



Do not finish this surface with roller or burnisher. This diameter to be measured when axle is cold. Axle to be round within 0.001 of an inch. Taper not to exceed 0.001 per foot on diameter.

18035

Fig. 26 – Axle Journal For .020" Undersized Inner Race

TAPERED CARTRIDGE TYPE ROLLER BEARING

The minimum axle journal diameter is governed by the interference fit between the axle journal diameter and the bore of the bearing cone assembly.

A new axle journal diameter is 6.1915" to 6.1905" for the Class F bearing and 7.004" to 7.003" for the Class G bearing. This provides an interference fit of .004" maximum to .0015" minimum.

The new minimum interference fit of .0015" is also the service limit.

FRICTION BEARING

Friction bearing axle journals that have been worn mechanically may be reduced to not less than 6.250" diameter and a 3/4" minimum fillet radius must be maintained. All tool marks must be polished from the fillet. If an axle journal has worn to 6.215" diameter, the axle should be scrapped.

SUPPORT BEARING AREA

The maximum axle diameter reduction at the support bearings should not exceed 1/16" due to the axle loading on some locomotives. However, 3/32" and 1/8" undersize support bearings are available for railroads wanting bearings below the recommended minimum undersize of 1/16".

WHEEL SEAT AREA

Wheel seats may be ground undersize to the limits given in the Service Data. However, at gear end, the transition between the wheel seat and the gear seat must be a smooth radius and the wheel seat length should be increased from 1/16" minimum to 5/32" maximum to ensure that the undersized wheel bore does not contact the shoulder.

JOURNAL BEARING INNER RACE

OIL LUBRICATED STRAIGHT ROLLER BEARING

Induction heater 8254202 is recommended for removing and installing the journal bearing inner race.